Electricity

Science • Chapter 11 • *Chapter Summary & Study Guide for Daksh*

Electricity is one of the most important forms of energy in modern life — driving everything from a tube light to a metro train. This chapter develops the basic physics: **electric current**, **potential difference**, **resistance** (Ohm's law), how resistors combine in series and in parallel, and the **heating effect** of current that powers our bulbs and heaters.

Current and potential difference

- **Electric current (I)** — flow of charge per unit time. Unit: **ampere (A)**. 1 A = 1 coulomb/second. Conventional current flows from + to – outside a cell (electrons move opposite).
- **Potential difference (V)** between two points = work done per unit charge moved between them. Unit: **volt (V)**. 1 V = 1 J/C. A cell or battery is a source of potential difference.
- Measured by: **ammeter** (in series, low resistance) for current; **voltmeter** (in parallel, high resistance) for potential difference.

Ohm's law

At constant temperature, the current through a metallic conductor is directly proportional to the potential difference applied across its ends:

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          V = I R
where R is the resistance, measured in OHMS (Ω).
1 Ω = 1 V/A.

Resistance opposes current. It depends on:
  - length L (longer wire -> more resistance)
  - area A (thicker wire -> less resistance)
  - material (resistivity ρ)
  - temperature (in metals, R increases with T)

      R = ρ L / A
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Combination of resistors

- **Series:** resistors joined end-to-end. Same current through all. R_s = $R_1 + R_2 + R_3 + \dots$ (always more than the largest individual).
- **Parallel:** resistors joined between two common points. Same potential difference across all. 1/R_p = $1/R_1 + 1/R_2 + \dots$ (always less than the smallest individual).

In **household wiring**, lights, fans, appliances are connected in **parallel**. Why? Each gets the full supply voltage (220 V); each can be switched on/off independently; if one fails, others keep working.

Heating effect of current

When current flows through a resistor, electrical energy is converted into heat. The amount in time t is:

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    H = I^2 R t   =  V I t   =  (V^2 / R) t

    P = V I  =  I^2 R  =  V^2 / R    (power, in watts)
    1 unit of electrical energy = 1 kWh = 3.6 × 10^6 J
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Practical applications: incandescent bulb (tungsten filament — high resistivity and very high melting point ~3380 °C — glows brightly when heated white-hot), electric heater, geyser, iron, toaster. **Fuses** use this effect deliberately: a thin alloy (Sn–Pb) wire of low melting point melts when current exceeds the rated value, breaking the circuit and saving appliances/wiring.

Practical pointers and safety

• Power rating of an appliance (e.g. 100 W, 220 V) tells you that at 220 V it draws ~0.45 A.

• Three-pin plugs include an **earth wire** — connects metal body to ground; if a fault occurs and the body becomes live, current escapes safely to earth and protects the user.

• Energy bill is in kilowatt-hours (units), not watts. 1 kWh = 1000 W operated for 1 hour.

★ MUST-REMEMBER POINTS

• $I = Q/t$ (current is charge per second).

• Ohm's law: $V = IR$. R in ohms (Ω). $R = \rho L/A$.

• Series: $R_s = R_1 + R_2 + \ldots$; Parallel: $1/R_p = 1/R_1 + 1/R_2 + \ldots$

• Household wiring is parallel; each appliance gets full voltage; independent control.

• Heat: $H = I^2Rt$; Power: $P = VI = I^2R = V^2/R$.

• 1 unit of electricity = 1 kWh = 3.6×10^6 J.

• Fuse melts to break the circuit at excess current — uses heating effect deliberately.

— End of Summary —